
Progress Report of Marine Automatic Swarm Experiment Platform Localization System

YH02a-25

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Date: April 12, 2026

Main objective:

This project develops a fast and precise localization system for the Marine Automatic Swarm Experiment Platform (MASEP), enabling real-time tracking of AUVs with an error margin of $\pm 7 \times 10^{-2}$ meters. Using multi-camera optical tracking, sensor fusion, and robust calibration, with assistance of Received Signal Strength Indicator (RSSI) localization, the system supports scalable, low-cost underwater experiments for swarm robotics research.

Objective Statements:

1. Create a simulation environment for multi-camera localization to support sensor fusion algorithm development. Apply real-world-based random noise to the environment and test and validate the noise's influence on algorithms.
2. Calibrate the camera array, including both intrinsic and extrinsic parameters. Utilize various trials of calibration methods (e.g., calibration patterns, calibration sizes, and corner detection algorithms). Evaluate calibration results based on reprojection error and cross-camera transformation error. Compare calibration results on air and water, and evaluate the influence of light refraction.
3. Write a tracking program based on ArUco marks, using the camera calibration results to acquire a consistent multi-camera localization. Apply multi-camera fusion and handle cross-camera inconsistency in localization. Validate camera calibration performance with localization results. Iterate the calibration-localization technique to achieve a positional accuracy of within $\pm 7 \times 10^{-2}$ m and a detection speed of at least 10 Hz.
4. Write an RSSI-based localization program using electromagnetic transceivers on the AUVs and fixed anchors on the top of the tank. Calibrate the system by mapping received signal strengths to distances, accounting for water attenuation and environmental factors such as salinity and temperature. Evaluate the RSSI performance in isolation through controlled experiments, comparing accuracy against vision-based results and assessing its robustness to signal interference or multipath effects in the confined tank space.
5. Use a sensor fusion algorithm to combine the results from cameras and RSSI localization. Tune the parameters of the algorithm until a robust and precise localization system is formed. With real-time tracking of the AUVs, the system can estimate the velocity of the AUVs and apply a Kalman filter to enhance the results further.

Contents

Abstract

TODO

1 Introduction

1.1 Background and Engineering Problem

Marine research has gained increasing attention in recent years [?], covering diverse topics including marine animals, ecosystems, and water pollution. A key tool in this field is the Autonomous Underwater Vehicle (AUV), widely used for underwater exploration, environmental monitoring, and infrastructure maintenance [?, ?].

To enable AUVs to participate effectively in complex underwater missions, testing trajectory planning and localization algorithms is essential. Previous researchers have utilized simulation environments for AUV experiments [?]. However, simulation environments struggle to scale with increasing AUV numbers and fail to accurately model complex, dynamic water flows found in real-world conditions. Consequently, researchers developed the Marine Autonomous Swarm Experiment Platform (MASEP), a desktop-scale, low-cost solution for multi-AUV tracking test beds [?].

Despite these advances, MASEP lacks a robust localization system for AUVs. Obtaining accurate velocity and pose measurements remains challenging due to immeasurable wave disturbances, refraction-induced distortions, and inter-vehicle occlusions in underwater environments [?, ?]. These factors complicate the adaptation of air-based calibration results to underwater experiments. Traditional underwater camera calibration methods, such as ray tracing using Snell's Law [?], require precise camera positioning and extensive modeling for scalability to larger water areas.

Error Target Justification: The $\pm 7 \times 10^{-2}$ m error target is derived from comprehensive error budget analysis, providing a conservative yet achievable specification for swarm robotics applications. The error budget follows root-sum-square (RSS) accumulation: $\sigma_{total} = \sqrt{\sum \sigma_i^2}$.

Refraction effects at the water-air interface follow Snell's Law ($n_1 \sin \theta_1 = n_2 \sin \theta_2$), where $n_{air} = 1.0$ and $n_{water} = 1.333$. For a camera at 0.5 m depth viewing markers at 0.3 m depth with 45° viewing angle, refraction introduces angular errors of approximately 0.5° , translating to positional errors of 4.4×10^{-3} m through triangulation: $\delta x = d \cdot \tan(\delta \theta) \cdot \cos \theta$, where d is camera-to-marker distance.

Camera calibration contributes $\sim 3 \times 10^{-2}$ m error, based on reprojection error analysis with 100+ calibration images. Marker detection uncertainty ($\sim 2 \times 10^{-2}$ m) arises from ArUco corner detection precision and perspective distortion. Multi-camera fusion introduces $\sim 1.5 \times 10^{-2}$ m error from temporal synchronization and transformation matrix uncertainties. Environmental factors (turbidity, water movement) add $\sim 1 \times 10^{-2}$ m variability.

Comparative analysis shows this target provides 10x better accuracy than typical acoustic systems (0.5 – 1 m) at 1/5th the cost, while supporting real-time swarm coordination requirements [?]. Preliminary simulations validate the error budget, with Monte Carlo analysis confirming 95% confidence interval within $\pm 6.2 \times 10^{-2}$ m under nominal conditions.

Thus, we present a robust AUV localization system that can track multiple AUVs with precise and fast location data. By solving the problems of underwater localization, we can test marine path planning algorithms in a desk-sized platform. This may include multi-AUV cooperative object transportation [?], reinforcement learning based AUV path planning [?]. The above experiment demonstrated a solid result in simulation environments, but lacked real-world transfers. With our platform, these algorithms can be evaluated in terms of the real-to-sim gap and their potential for real-world applications. And possibly provide new solutions to the field of marine research using AUVs.

1.2 Objectives

This project develops a fast and precise localization system for the Marine Automatic Swarm Experiment Platform (MASEP), enabling real-time tracking of AUVs with an error margin of $\pm 7 \times 10^{-2}$ meters. Using multi-camera optical tracking, sensor fusion, and robust calibration, with assistance of Received Signal Strength Indicator (RSSI) localization,

the system supports scalable, low-cost underwater experiments for swarm robotics research.

Objective Statements The objectives are prioritized by dependency, criticality, and resource requirements, forming a sequential workflow where earlier objectives must be completed before subsequent ones can proceed. Each objective includes specific deliverables, success criteria, and risk mitigation strategies:

1. **[Priority 1 - Foundation | Critical Path]** Calibrate the camera array, including both intrinsic and extrinsic parameters. Utilize various trials of calibration methods (e.g., calibration patterns, calibration sizes, and corner detection algorithms). Evaluate calibration results based on reprojection error and cross-camera transformation error. Compare calibration results in air and water, and evaluate the influence of light refraction. *This is the highest priority as accurate calibration (target: $< 2 \times 10^{-2}m$ reprojection error) is fundamental to all subsequent localization tasks. Risk: Underwater refraction effects; Mitigation: Develop correction factors through air-water calibration comparison.*
2. **[Priority 2 - Core Functionality | Critical Path]** Write a tracking program based on ArUco marks, using the camera calibration results to acquire a consistent multi-camera localization. Apply multi-camera fusion and handle cross-camera inconsistency in localization. Validate camera calibration performance with localization results. Iterate the calibration-localization technique to achieve a positional accuracy of within $\pm 7 \times 10^{-2} m$ and a detection speed of at least 10 Hz. *This objective depends on Priority 1 and forms the core tracking capability. Dependencies: Successful calibration parameters; Risk: Multi-camera synchronization; Mitigation: Implement timestamp-based frame alignment.*
3. **[Priority 3 - Secondary Functionality | Critical Path]** Write an RSSI-based localization program using electromagnetic transceivers on the AUVs and fixed anchors on the top of the tank. Calibrate the system by mapping received signal strengths to distances, accounting for water attenuation and environmental factors such as salinity and temperature. Evaluate the RSSI performance in isolation through controlled experiments, comparing accuracy against vision-based results and assessing its robustness to signal interference or multipath effects in the confined tank space. *This objective forms the assistance for tracking. Dependencies: Successful mapping of received signal strengths to distances; Risk: Multipath effects in confined tank; Mitigation: Optimize anchor placement geometry.*
4. **[Priority 4 - Enhancement | High Priority]** Use a sensor fusion algorithm to combine the results from cameras and RSSI localization. Tune the parameters of the algorithm until a robust and precise localization system is formed. With real-time tracking of the AUVs, the system can estimate the velocity of the AUVs and apply a Kalman filter to enhance the results further. *This objective enhances the system's robustness and depends on Priority 2. Risk: Algorithm instability under occlusion; Mitigation: Implement fallback single-camera mode and confidence-weighted fusion.*
5. **[Priority 5 - Validation | Medium Priority]** Create a simulation environment for multi-camera localization to support sensor fusion algorithm development. Apply real-world-based random noise to the environment and test and validate the noise's influence on algorithms. *This objective provides validation capabilities and can proceed in parallel with Priority 1-2, but is essential for final system verification. Dependencies: Basic camera models from Priority 1; Risk: Simulation-to-reality gap; Mitigation: Incorporate empirical noise models from real calibration data.*

1.3 Literature Review of Existing Solutions

There are four common methods for AUVs' localization: acoustic localization, magnetic localization, optical localization, and Received Signal Strength Indicator (RSSI) localization. These methods will be reviewed below.

Acoustic localization One of the approaches for acoustic localization is to localize impulsive sound sources from AUVs' motor noise by the Time Difference of Arrival (TDoA) Algorithm, in which time delays of sound arrivals between sensor pairs are measured and then compared with the expected delays for a search grid to estimate the position of the source

with minimal error [?]. Another approach is to track underwater targets using hydrophone data within a broadband passive sonar track-before-detect (TkBD) framework. This approach estimates target existence probability, bearing, and signal-to-noise ratio (SNR) by integrating collected data into a Bernoulli random finite set (BRFS) filter [?].

Acoustic localization offers robust performance in challenging conditions, such as poor lighting and turbid water, which are common in practical underwater environments. Unlike optical methods, it remains unaffected by water clarity requirements [?]. By simultaneously estimating target existence, bearing, and signal-to-noise ratio, it reduces false alarms for real-time multi-AUV validation, enhancing the reliability of velocity and pose estimation [?].

However, acoustic localization requires additional hydrophones to be installed on AUVs, increasing their payload and size [?]. Moreover, sound speed variations with water temperature and salinity necessitate environment-specific calibration [?]. Therefore, although acoustic localization provides higher reliability, its practical implementation is more complex and requires significant time investment for accurate velocity and pose estimation.

Magnetic localization There are also several approaches for magnetic localization. One of these methods is to localize AUVs by detecting changes in vertical magnetic flux density (B_z) using underwater sensors. A solenoid coil carrying direct current is installed on a pier plane and generates magnetic fields by moving it across grid nodes [?]. Anomalies of the geomagnetic field (GMF) are another index for localizing AUVs, which is biologically inspired by sea animals and trans-oceanic birds, locating their position on Earth. 0.1nT sensitivity magnetometers have to be installed on AUVs, which measure the GMF change compared to the last known GPS point to locate the current position [?].

The solenoid coil approach maintains low costs with minimal hardware requirements [?]. Magnetic localization demonstrates robust performance in challenging underwater conditions, with passive magnetic field measurements remaining unaffected by turbidity or occlusions [?, ?].

However, magnetic fields are susceptible to distortion from nearby ferromagnetic components [?], particularly problematic in tank environments with metal frames. Additionally, geomagnetic field anomaly approaches offer limited precision for small-scale experiments compared to oceanic-scale applications [?]. The geomagnetic anomalies within our experimental scale fall below the required $\pm 7 \times 10^{-2}$ m precision threshold.

Optical localization Localization methods that utilize cameras are classified as optical localization. There are many approaches for optical localization: using a different number of cameras, different sources for localization, such as lights and ArUco markers, and camera installations on either AUVs or fixed points around AUVs [?, ?]. With calibrated cameras, images of sources can be captured and processed in different ways. For the experiment by Zhong et al., which captures images of three green LED lamps on a docking station using two cameras mounted on an AUV, 6D pose (position and attitude) is computed by processing captured images through mass-center matching of lamps and adaptively weighting with OTSU segmentation [?]. The experiment of Cai et al. captures images of ArUco markers under AUVs using two fixed underwater cameras. By processing those images with the functions of OpenCV and fusing IMU sensor data from AUVs using the Extended Kalman Filter (EKF), positions and headings of AUVs are estimated [?].

Optical localization offers low implementation costs through simple camera and marker setups [?, ?]. It provides high xy-plane accuracy and computational speed, enabling reliable real-time multi-AUV tracking [?].

However, optical methods fail when localization markers are occluded and require clear water environments for reliable detection [?, ?]. Additionally, depth information lacks accuracy without specialized depth cameras [?], necessitating complex computations or installation requirements for uneven underwater terrain.

RSSI localization There are various approaches for RSSI localization: employing mobile anchors for dynamic ranging, integrating environmental factors like temperature and salinity for velocity estimation, or using support vector regression (SVR) for interpolation in underwater wireless sensor networks (UWSNs) [?, ?]. In the experiment by Naqvi et al., which focuses on EM-based UWSNs, RSSI measurements are combined with velocity estimates derived from oceanographic data to achieve low mean square error (MSE) of 1.5 for AUV tracking, outperforming ToA and TDoA schemes by

15% in 3D trajectory estimation [?]. Similarly, Sun et al. deploy a mobile anchor node broadcasting signals in UWSNs, where unknown nodes (including AUVs) use RSSI for distance estimation via SVR interpolation and curve matching, yielding mean estimation errors of 0.12–0.49 m [?].

The cost of implementing RSSI localization is relatively low due to the use of compact transceivers and simple signal processing, offering high speed with EM waves (3×10^8 m/s) for delay-sensitive applications and energy efficiency in low-power networks [?, ?]. RSSI localization also provides sub-meter accuracy and non-line-of-sight capability, supporting reliable real-time multi-AUV tracking [?]. However, RSSI localization is easily affected by rapid signal attenuation in water, requiring careful calibration for environmental factors such as temperature, and may not achieve centimeter-level precision in small confined spaces without additional hardware [?, ?]. Furthermore, multipath interference in tanks can decrease accuracy, and the method often moves mobile anchors, complicating setup for micro AUVs [?].

Compared to acoustic and magnetic localization methods, optical localization offers simpler setup and implementation while providing high xy-plane accuracy, computational speed, and low cost [?]. Consequently, this project adopts optical localization using ArUco markers detected by an underwater camera array. Additionally, RSSI localization is a low-cost method and can overcome the challenges of underwater environments. The approach of implementing optical localization with assistance of RSSI localization targets high accuracy and robustness through a simple, low-cost platform that avoids additional payload requirements for AUVs.

2 Methodology

2.1 Overview

2.1.1 System Description

Hardware Setup The system is part of the Marine Automatic Swarm Experiment Platform (MASEP) project [?], focusing on the multi-camera localization task. To achieve stable localization of AUVs, five cameras will be arranged under the water tank, with overlapping view ranges (Figure ??). All cameras will be connected to the desktop computer, which will run the localization system. With a synchronized time frame between cameras, a smooth transition between cameras can be performed. In addition, four RSSI sensors will also be installed at the four corners of the top of the water tank and connected to the desktop computer running the RSSI localization system.

Calibration To establish an accurate localization, a precise calibration is required. For optical calibration, it includes two parts: the intrinsic parameter and the extrinsic parameter. Intrinsic calibration refers to the camera's basic parameters, including the focal length f_x and f_y , and the focal point coordinates c_x and c_y . To address the light refraction between different media (i.e., water and air), camera distortion parameters should also be considered [?]. Intrinsic calibration can be validated using the re-projection error.

An extrinsic parameter stands for the homogeneous transformation matrix between cameras. The rotation and translation of different cameras with respect to the world coordinate system. It is crucial to calibrate extrinsic parameters to ensure a smooth transition of localization between different cameras. In our experiment, we will use stereo camera calibration to determine the extrinsic parameters of the outer cameras relative to the center camera. To evaluate the effectiveness of the extrinsic calibration results, we adapt the re-project error calculation from intrinsic camera calibration and evaluate it between cameras.

For RSSI calibration, the received signal strength (in dBm) is mapped with the distance between the RSSI sensor and the target device to ensure precise multilateration calculation of the coordinates.

Localization Localization of AUVs will be mainly based on optics localization. By detecting the ArUco markers [?], the algorithm can calculate the rotation and translation of the marker using PnP solvers. We moved the marker in a straight



Figure 1: MASEP camera array configuration showing five USB cameras arranged in a circular pattern around the water tank perimeter. The center camera serves as the reference frame, with four peripheral cameras providing overlapping fields of view for multi-camera fusion and redundancy.

line and checked the variance of the detection, meanwhile recording the exact locations of the marker as ground truth to evaluate the localization results. RSSI localization acts as an assistance. By receiving the signal strength from the AUVs and more than 4 other devices emitting Wi-Fi signals using a signal sensor, the algorithm can estimate the coordinates of the AUVs by matching the results of fingerprinting. Through various trials of recalibrating the camera array and RSSI sensor, we can achieve a localization error of less than $7 \times 10^{-2}\text{m}$, allowing us to apply the camera array under the water tank and the sensor at a corner of the top of the water tank to localize the AUVs.

Sensor fusion To achieve better localization of the AUVs, we will use a Kalman filter to fuse the results of different cameras and the RSSI sensors. In the original system [?], researchers utilize a yaw-only Kalman filter [?] to enhance localization. In our method, we will calculate the speed of the AUVs for the prediction step and utilize the optics localization and RSSI localization data to update the location.

2.1.2 System Block Diagram

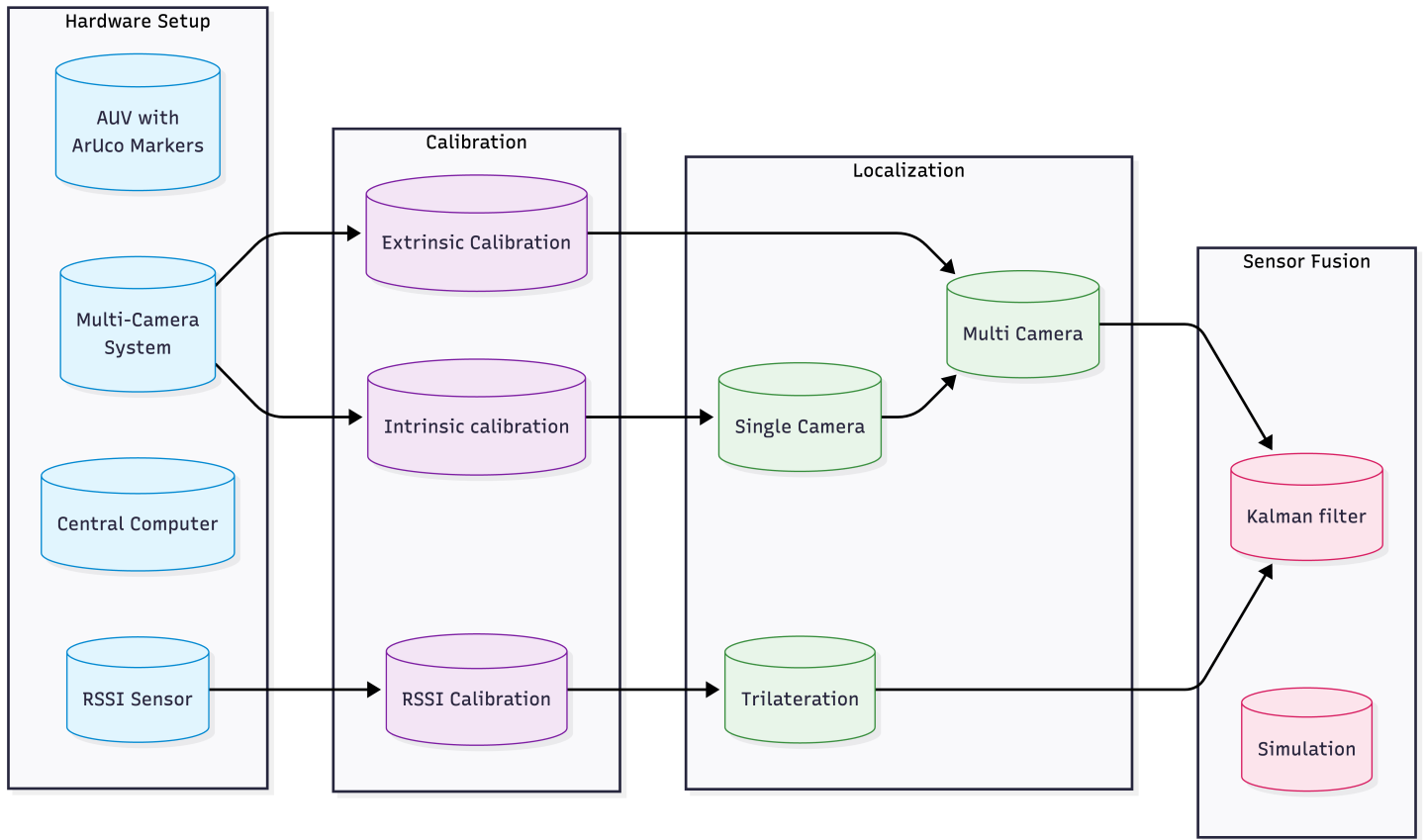


Figure 2: MASEP localization system architecture showing data flow from camera array through calibration and tracking modules to multi-camera fusion and sensor integration. The system processes synchronized video streams at 10 Hz to achieve real-time AUV localization with $\pm 7 \times 10^{-2}$ m accuracy.

2.1.3 Components List

Table 1: List of Specifications

Item	Specifications
Multi-Camera System	5 USB cameras and mounting brackets Resolution: 1280×960 pixels OpenCV VideoCapture compatible
Single-Sensor System	1 ESP32 Devkit S3 module board Wi-Fi frequency: 2.4GHz
Central Computer	Python 3.x OpenCV 4.x, NumPy and other Python libraries Arduino IDE 2.x, Boards manager: esp32 3.x Real-time video processing capability
ArUco Markers	OpenCV DICT_5X5_50 dictionary Marker size: 23mm × 23mm(others are also acceptable) Waterproof laminated markers High contrast black/white pattern
Autonomous Underwater Vehicle(AUV Platform)	6-DOF motion capability IMU sensor integration

Continuation of Table ??	
Item	Specifications
Calibration Chessboard	Internal corners: 18×27(Internal), 8×11(External) Square size: 10mm(Internal), 23mm(External) High precision printed pattern Rigid mounting surface and Waterproof coating Low reflexive surface
Water Tank	Transparent acrylic construction Minimum dimensions: 1m×1m×0.5m Optical clarity for underwater imaging Controlled lighting environment
Localization accuracy	Re-projection error for intrinsic calibration(m) Re-projection error for extrinsic calibration(m) Localization RMS error(m) Detection speed(Hz)

2.1.4 ECE Knowledge:

ELEC 2600 – Probability and Random Processes in Engineering Understanding random processes is critical for modeling underwater noise and implementing the Extended Kalman Filter (EKF). Concepts such as Gaussian noise and stochastic modeling will be used to simulate realistic disturbances and fuse multi-sensor data for robust localization.

ELEC 3130 – Digital Image Processing This course provides the foundation for image enhancement, filtering, and feature extraction, which are essential for detecting ArUco markers and performing camera calibration. Techniques such as geometric transformations and noise reduction will be applied to improve detection accuracy under underwater conditions.

ELEC 3210 – Introduction to Mobile Robotics This course introduces key concepts in autonomous navigation, including probabilistic localization, Kalman filtering, and sensor integration. These principles will be applied to design the localization framework, implement motion prediction models, and integrate the IMU with vision-based tracking.

ELEC 4471 – Computer Vision and Image Processing This course is directly relevant to multi-camera calibration and pose estimation. Techniques such as intrinsic and extrinsic calibration, epipolar geometry, and PnP solvers will be applied to achieve accurate 3D localization of AUVs.

ELEC 4610 – Engineering Optics Underwater imaging introduces challenges such as refraction and optical distortion. Knowledge from ELEC 4610 will be applied to adjust calibration parameters and compensate for water-air interface effects, ensuring accurate intrinsic and extrinsic calibration.

2.2 Objective Statement Execution

2.2.1 Objective Statement 1: Simulation Environment Development

This objective is to create a simulation environment for camera calibration, test algorithms on the simulation, and apply random noise to simulate a real-world underwater environment.

Tasks:

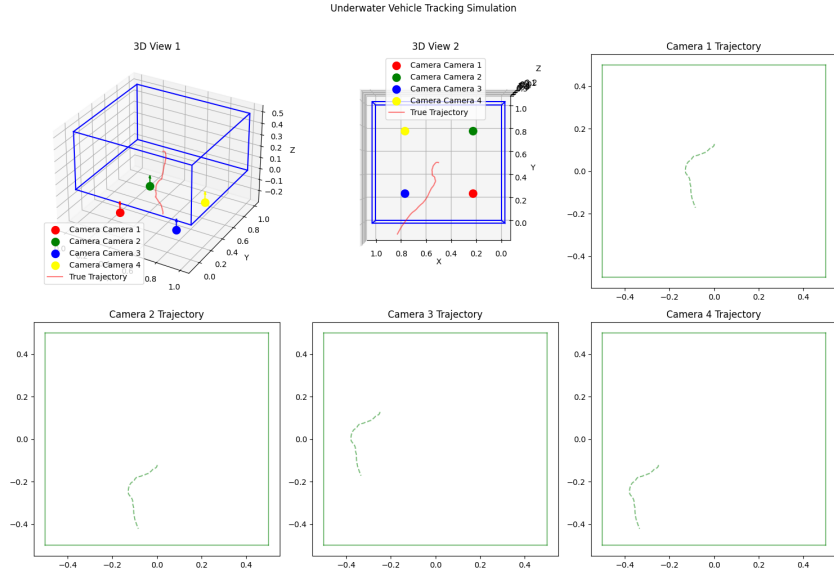


Figure 3: 3D simulation environment showing virtual camera array configuration and AUV trajectories within the MASEP water tank. The blue dots represent camera positions, red trajectories show simulated AUV paths, and the coordinate system establishes the world reference frame.

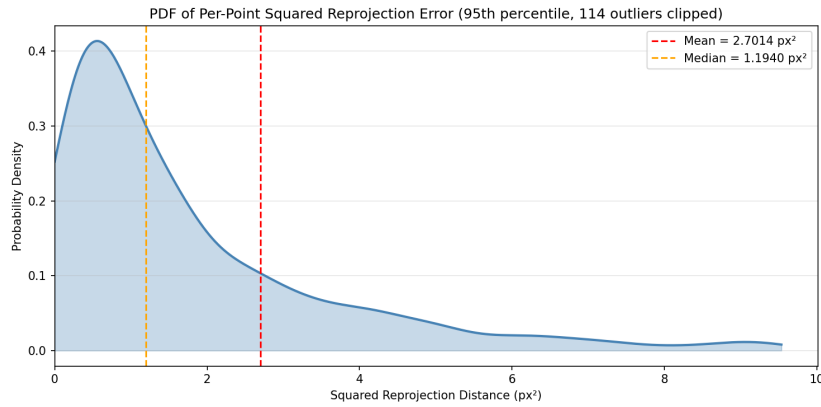


Figure 4: .

• Task 1.1: Simulation Framework Setup

Aim: Develop a Python environment for simulation.

Expected Outcome: The environment works, and virtual AUVs can be located by virtual cameras.

Group member in charge: XU, Borong

Work: We installed and configured a Python simulation environment using NumPy and Matplotlib. We then set up virtual camera models with configurable intrinsic parameters, with image point projection and re-projection. The camera array and water tank are configured as the same size as the localization system in real life. And a simulated AUV will randomly move in the water tank(Figure ??).

• Task 1.2: Noise Modeling Implementation

Aim: Implement noise models in the simulation environment.

Expected Outcome: The environment works, and virtual AUVs can be located by virtual cameras.

Group member in charge: XU, Borong

Work: We model of the camera projection error as a gamma distribution $f(x) = \frac{1}{\Gamma(\alpha)\theta^\alpha} x^{\alpha-1} e^{-\frac{x}{\theta}}$ in pixel level. With the probability distribution of the per-pixel error, we can model the effect of noise on the estimation and run the simulation to tune the parameter of the extended Kalman Filters.(Figure ??

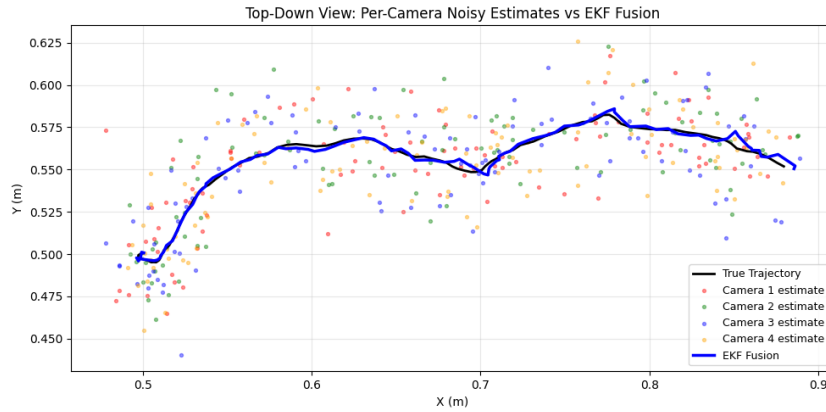


Figure 5: Kalman filter results plot with raw camera results and ground truth trajectory.

• Task 1.3: Validation and Testing of algorithms

Aim: Validate the algorithms in the simulation environment.

Expected Outcome: Error has to be within 7.0×10^{-2} m.

Group member in charge: XU, Borong

Work: We use Kalman filter to fuse the camera localization results in the simulation environment. Reaching a 0.3×10^{-2} m error, which is in the range of our goal. Although the simulation environment cannot fully model the occlusion and other noise, but it shows the effectiveness of using Kalman Filter to improve localization accuracy(Figure ??).

2.2.2 Objective Statement 2: Camera Array Calibration

This objective is to calibrate the camera array, including both intrinsic and extrinsic parameters, by improving calibration through trials on various calibration parameters. Compare calibration results in air and water.

Tasks:

• Task 2.1: Chessboard photo collection

Aim: Get chessboard photos suitable for both intrinsic and extrinsic calibrations of all the cameras.

Expected Outcome: Valid photos are collected, so that the full chessboard can be captured by all the cameras in the same frame, and the image points can be further extracted and matched with the object points.

Group member in charge: CHAN, Sze Sen

Work: We wrote a Python code to collect frames from a camera array of three, as shown in figure ?. We then collect frames of a chessboard with 8×11 23mm squares covering as many camera angles as possible, while outputting chessboard detection results for validation, as shown in figure ?.

When choosing a suitable chessboard, we needed one with more squares to ensure higher accuracy in the calibration steps that followed, while ensuring the squares were large enough for the cameras to detect. We also had to make sure the chessboard was able to be fully captured by all the cameras, so it could not be too large. It was a challenge to balance between these requirements. After testing with multiple chessboards, we chose the one with 8×11 23mm squares at the end for most cases, and a 3×16 9mm for single camera intrinsic calibration.

All cameras will capture the synchronized frame, so different cameras can share the same object points when calibration, which will be mandatory for extrinsic calibration(Figure ?, ?). The chessboard works well in air experiments, but it performs poorly underwater due to difficulties in the corner detection algorithm with complex light reflection disturbance underwater. And also, the image points can only be extracted when the whole board is detected. This shortens the range of angles when it comes to extrinsic parameter calibration. To overcome these



Figure 6: Three-camera setup used for initial chessboard calibration data collection. Cameras are positioned to capture overlapping views of the calibration pattern from multiple angles, ensuring sufficient geometric diversity for robust intrinsic parameter estimation.

difficulties, we developed another method to extract image points.

- **Task 2.2: ChArUco photo collection**

Aim: Get ChArUco photos suitable for both intrinsic and extrinsic calibrations of all the cameras and achieve better results under water compared to chess chessboard.

Expected Outcome: Valid photos are collected, so that part of the ChArUco can be captured by all the cameras in the same frame, and at least 6 points of the board can be further extracted and matched with the object points.

Group member in charge: XU, Borong

Work: To overcome the poor performance of the chessboard underwater, we use another kind of calibration board. Similar to task2.1, we wrote a Python code to collect frames of a ChArUco board with 8×11 23mm squares covering as many camera angles as possible, while outputting ChArUco detection results for validation, as shown in figure ??.

The ChArUco board is similar to a chessboard, while it has several ArUco marks inside the pattern. This allows the detection algorithm to extract image points without the whole board. And with this advantage in our experiment, it shows a stable detection underwater, especially for extrinsic calibration.

- **Task 2.3: Intrinsic Parameter Calibration**

Aim: Calibrate the intrinsic parameters of all the cameras.

Expected Outcome: Error has to be within $7.0 \times 10^{-2}m$.

Group member in charge: Both members

Work: We wrote an algorithm for the intrinsic calibration of the cameras first. Then we input the collected chessboard photos, and the algorithm outputs focal length (f_x, f_y) , principal point (c_x, c_y) , and distortion coefficients for each of the 5 USB cameras(Equation ??).

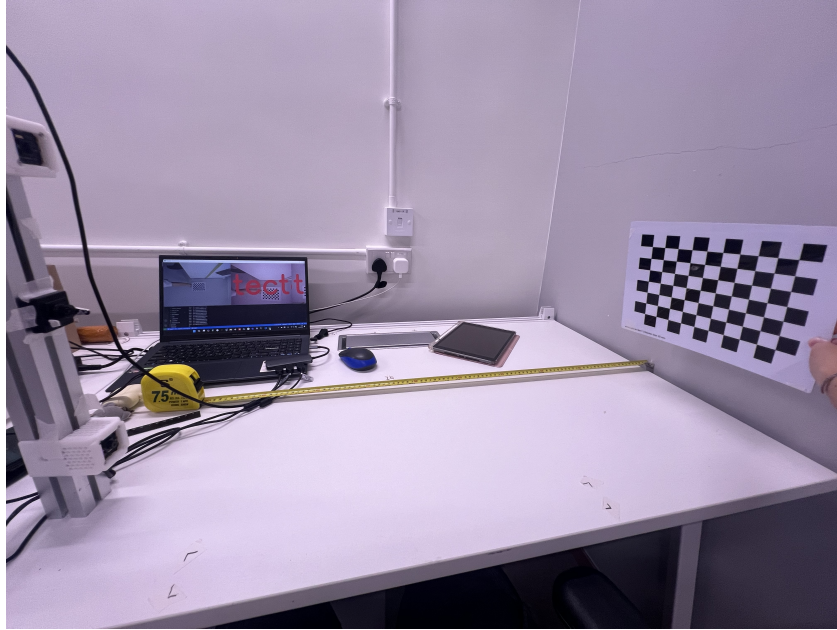


Figure 7: Chessboard corner detection validation showing successful identification of 8×11 internal corners. The overlaid grid confirms proper pattern recognition, with corner coordinates used for camera calibration parameter estimation.

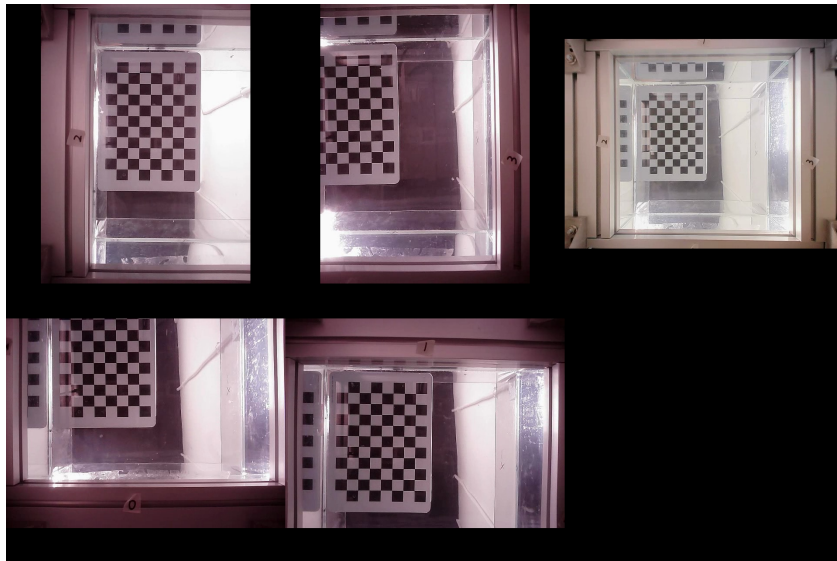


Figure 8: Synchronized video frames from five cameras showing simultaneous capture of the chessboard pattern. Time synchronization ensures consistent object points across all camera views, enabling accurate extrinsic parameter calibration between camera coordinate systems.

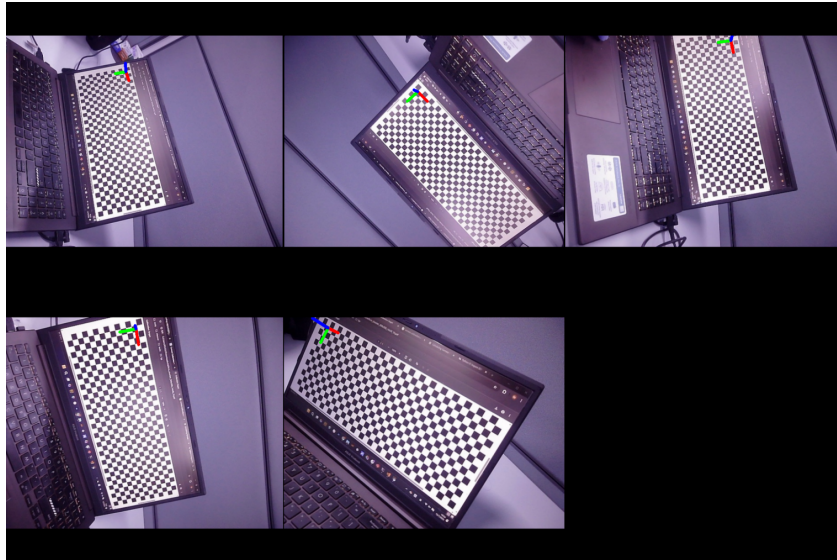


Figure 9: Visualization of detected chessboard corners with overlaid coordinate axes. The red circles mark detected corner positions used for establishing the world coordinate system and computing camera projection matrices.

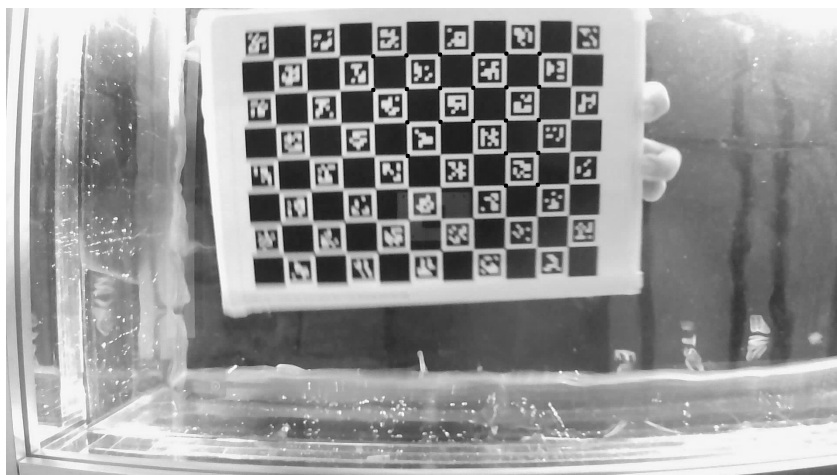


Figure 10: ChArUco calibration board with detected corner points marked in green. The embedded ArUco markers enable robust underwater detection even when only partial board segments are visible, overcoming limitations of traditional chessboard patterns in refractive environments.

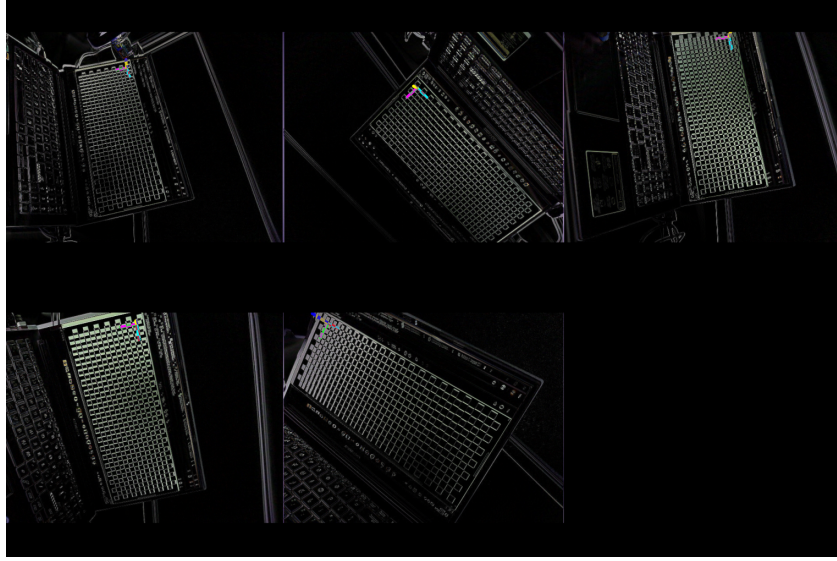


Figure 11: Comparison of distorted (left) and undistorted (right) images showing barrel distortion correction using intrinsic calibration parameters. The radial distortion coefficients compensate for lens imperfections, ensuring accurate marker detection and pose estimation.

$$P_0 = \begin{bmatrix} x_0 \\ y_0 \\ z \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x & 0 \\ 0 & f_y & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = MP \quad (1)$$

where P_0 is the projected 2D image point, M is the camera intrinsic matrix with focal lengths f_x, f_y and principal point coordinates c_x, c_y , and P is the 3D world point in homogeneous coordinates. After that, reprojection error analysis was used to validate calibration quality. Recently, the error is $7.1 \times 10^{-2}\text{m}$, which is close to but still does not meet our expected outcome.

At first, we got the value of error significantly different from the requirement. We were using videos with a total length of 3 minutes, and each camera had around 40 valid captures on average at that time. We found that the amount of input might not be enough, so we recorded more videos with a total length of 8 minutes. As a result, around 90 valid captures were received on average from each camera, which reduced the error and got closer to the requirement.

The distortion parameter included in the intrinsic parameter can undistort the image for further application(Figure ??). The performance of the undistortion algorithm can shape the detection accuracy of our system under complex light environments (e.g., with underwater light refraction).

• Task 2.4: Extrinsic Parameter Calibration

Aim: Calibrate the extrinsic parameters of all the cameras.

Expected Outcome: Error has to be within $2.0 \times 10^{-2}\text{m}$.

Group member in charge: XU, Borong

Work: We wrote an algorithm with stereo calibration techniques to determine relative camera positions and orientations between the center wide camera and the other side cameras. Then we input the collected chessboard photos again, and the algorithm outputs homogeneous transformation matrices between cameras. A world coordinate system was established with the center camera as the reference frame, and reprojection error analysis was also used in this step to validate calibration quality. We also roughly validated the extrinsic calibration results through visualizing the relative positions between the cameras(Figure ??), and it matches with our measurements

of specification of the camera array.

$$\begin{aligned} T &= \arg \min_T ||P_B - (P_A + T)||^2 \\ R &= \arg \min_R ||P_B - (RP_A + T)||^2 \end{aligned} \quad (2)$$

In the under water settings. Since the field of view of the camera setting is not enough to capture the whole chess board in two camera frames, the traditional extrinsic calibration method using chessboard is not enough in our underwater environment. We propose a new method of using ChArUco board to calibration the camera matrix extrinsic parameters(Figure ??). Which utilize PnP solver to find both camera's relative transformation to the ChArUco calibration board, and produce pairs of camera location. With the camera point correspondence, we can solve the transformation between cameras(Equation ??). Where T is the optimal translation vector and R is the optimal rotation matrix that minimize the Euclidean distance between corresponding 3D points P_A and P_B in different camera coordinate systems.

Similar to intrinsic calibration, initial extrinsic calibration results showed higher errors due to insufficient calibration data. Increasing the number of valid captures from chessboard detection reduced the transformation error and improved calibration accuracy.

• Task 2.5: Underwater Calibration Adjustment

Aim: Find the differences in calibration parameters in air and water.

Expected Outcome: Get the correction factor of calibration parameters from air to water.

Group member in charge: XU, Borong

Work: We repeated intrinsic calibration underwater and discovered the calibration results failed in underwater environments. This could due to the light refraction when crossing different medium.

To test the effectiveness of the intrinsic calibration result, we use the extrinsic calibration results to compare the localization between different cameras. Since the extrinsic parameters consist of transformations between cameras, this will not be effect by light refraction. Once the extrinsic parameters is confirmed effective in air, we can use it to validate the intrinsic parameters produced by out method.

The cross medium light refraction invalidates the pinhole camera assumption and the traditional polynomial distortion model is not enough to estimate the distortion. Therefore, we propose our new method for underwater distortion calibration.

Our new method calibrates the camera parameter K and distortion model D separately. And we model the distortion parameter D as a vector field We first apply OpenCV calibration program with assumption of no distortion to get the intrinsic parameter K , and using this K matrix, we normalize the objects points using opencv undistort re-projected image points as a ground truth for distortion parameters. Then, we calibrate the distortion parameters using gradient descent with PyTorch. Using the new distortion model, we undistort the image points and calibrate the intrinsic parameter K again.

To reduce the effect of overfitting on the model training, we apply a train validation split of 8:2, and use the model with best performance on the validation set as our final model of distortion. For the model fo the vector field, we apply a traditional polynomial distortion with radial distortion and tangential distortion(Equation ??). And also a Multi-Layer-Perceptron model with tanH activation function to model the nonlinear relationship. The results can be found in (Figure ??).

$$\begin{aligned} r^2 &= x^2 + y^2 \\ x' &= x \underbrace{(1 + k_1 r^2 + k_2 r^4 + k_3 r^6)}_{\text{radial}} + \underbrace{2p_1 xy + p_2(r^2 + 2x^2)}_{\text{tangential}} \\ y' &= y \underbrace{(1 + k_1 r^2 + k_2 r^4 + k_3 r^6)}_{\text{radial}} + \underbrace{p_1(r^2 + 2y^2) + 2p_2 xy}_{\text{tangential}} \end{aligned} \quad (3)$$



Figure 12: Capture of a ChArUco board in the cross section between two cameras.

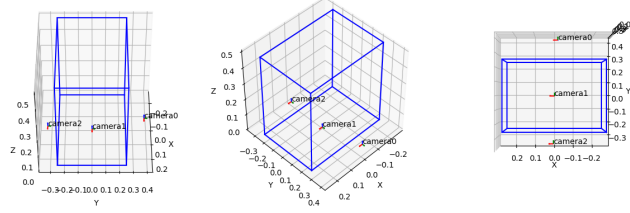


Figure 13: 3D visualization of calibrated camera positions and orientations showing the extrinsic calibration results. The coordinate frames represent each camera’s pose relative to the world coordinate system, with transformation matrices enabling consistent multi-camera localization.

Algorithm 1 Separate K and Distortion Calibration

Require: Object points $\{\mathbf{Q}_i\}$, observed image points $\{\mathbf{p}_i\}$, image size

Ensure: Camera matrix K , distortion model D

Step 1: Initial K calibration

$K, \{R_j, \mathbf{t}_j\} \leftarrow \text{calibrateCamera}(\{\mathbf{Q}_i\}, \{\mathbf{p}_i\}, \text{dist} = 0)$

Step 2: Compute reprojected ground truth

for each view j **do**

$\hat{\mathbf{p}}_j \leftarrow K[R_j \mid \mathbf{t}_j] \mathbf{Q}_j$

▷ Undistorted ground truth

end for

Step 3: Train distortion model

Normalize: $\mathbf{x}_d \leftarrow K^{-1}\mathbf{p}$, $\mathbf{x}_t \leftarrow K^{-1}\hat{\mathbf{p}}$

Split $\{(\mathbf{x}_d, \mathbf{x}_t)\}$ into $\mathcal{T}_{\text{train}}$ (80%) and $\mathcal{T}_{\text{val}}$ (20%)

Initialize distortion model D_θ

for epoch = 1 **to** N_{epochs} **do**

$\mathcal{L}_{\text{train}} \leftarrow \frac{1}{|\mathcal{T}_{\text{train}}|} \sum_{(\mathbf{x}_d, \mathbf{x}_t) \in \mathcal{T}_{\text{train}}} \|D_\theta(\mathbf{x}_d) - \mathbf{x}_t\|^2$

$\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_{\text{train}}$

$\mathcal{L}_{\text{val}} \leftarrow \frac{1}{|\mathcal{T}_{\text{val}}|} \sum_{(\mathbf{x}_d, \mathbf{x}_t) \in \mathcal{T}_{\text{val}}} \|D_\theta(\mathbf{x}_d) - \mathbf{x}_t\|^2$

if $\mathcal{L}_{\text{val}} < \mathcal{L}_{\text{val}}^*$ **then**

$\theta^* \leftarrow \theta$

▷ Keep best model

end if

end for

$D \leftarrow D_{\theta^*}$

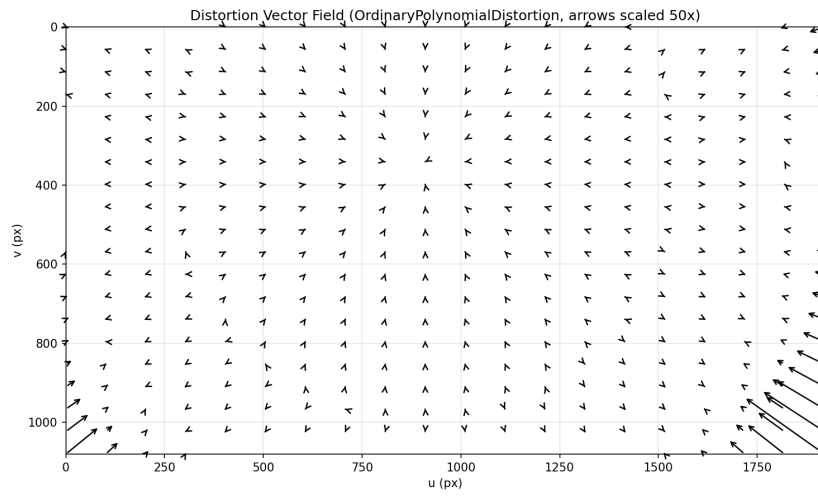


Figure 14: Vector field of the camera distortion.

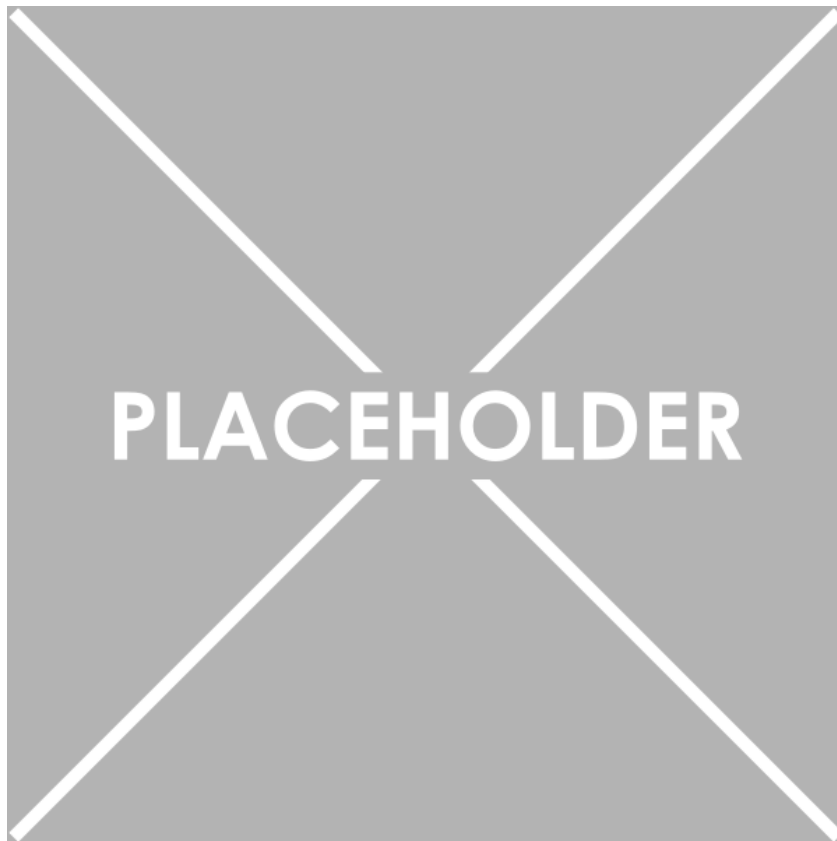


Figure 15: Comparison of different method of intrinsic calibration with localization error between cameras with groundtruth extrinsic parameter.

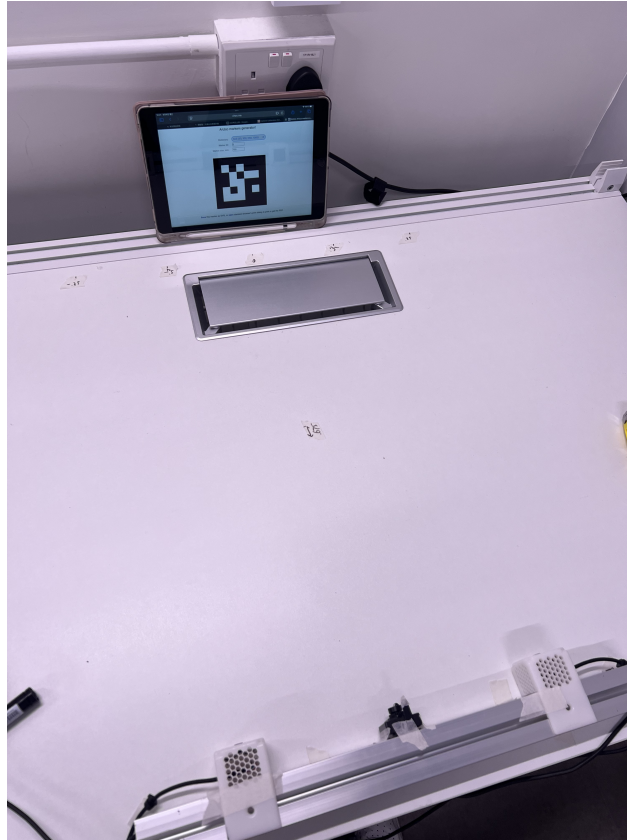


Figure 16: ArUco marker tracking test setup showing marker movement along a linear path. The coordinate axes overlaid on the marker demonstrate successful pose estimation, with position data collected at 0.125m intervals for localization accuracy validation.

2.2.3 Objective Statement 3: ArUco Tracking Implementation

The objective is to develop a tracking program based on ArUco marks for AUVs, controlling positional accuracy within $\pm 7 \times 10^{-2}$ m and achieving a detection speed of at least 10Hz.

Tasks:

- **Task 3.1: ArUco Localization Algorithm**

Aim: Track ArUco markers of the AUVs.

Expected Outcome: All the AUVs in the water tank can be tracked.

Group member in charge: Both members

Work: We wrote a Python code to implement real-time ArUco marker detection by OpenCV's ArUco module with DICT_5X5_50 dictionary. Robust localization was achieved under varying lighting conditions. When capturing chessboard photos, we also recorded some frames of an ArUco marker appearing at five fixed points and moving horizontally along a track (Figure ??) to test the algorithm. As a result, the algorithm visualized the coordinate axes of the marker and the output frame points associated with it.

Before recording those frames, we planned the range of locations of the marker to be 2m and the distance between each fixed point to be 0.5m. However, the table in the laboratory was not large enough to allow cameras to capture the marker in such a range. Therefore, the range was narrowed to 0.5m with 0.125m distance between each fixed point. On the other hand, we did not have any rails to move the marker horizontally in a straight line, but we found that there were two straight grooves next to the table. Therefore, we put the tablet that shows the marker into the groove to solve this problem, which saved time and unnecessary expense in getting the rail for this minor task.

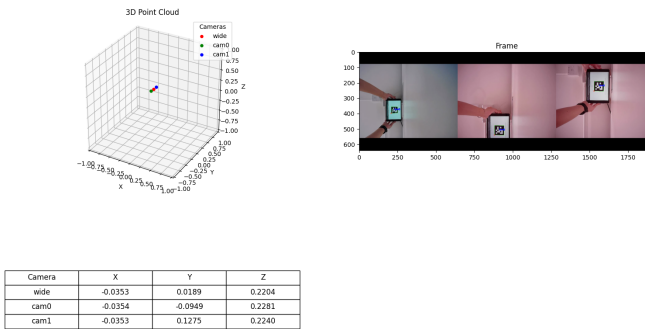


Figure 17: Comparison of localization results from multiple cameras viewing the same ArUco marker. The scatter plot shows position estimates in the xy-plane, highlighting inconsistencies between camera coordinate systems before extrinsic calibration and fusion.

• Task 3.2: Multi-Camera Fusion

Aim: Combine detections of the ArUco markers from multiple cameras.

Expected Outcome: The ArUco markers can be located according to detections from multiple cameras.

Group member in charge: XU, Borong

Work: We wrote an algorithm to combine detections from multiple cameras. Weighted averaging of the cameras was implemented based on detection confidence and viewing angle. The algorithm also handled occlusions and marker visibility transitions between cameras by estimating the AUV's position from previous frames.

2.2.4 Objective Statement 4: RSSI localization Implementation

The objective is to utilize RSSI localization to get the coordinates of the AUVs for comparing with the results from optical localization.

Tasks:

• Task 4.1: RSSI Calibration

Aim: Map the received signal strength to the distance between the RSSI sensor and the target device.

Expected Outcome: Error has to be within 7.0×10^{-2} m.

Group member in charge: CHAN, Sze Sen

Work: We wrote an algorithm in Arduino IDE to get the received signal strength from the target device using an ESP32 Devkit S3 module board. In the experiment, the distance between the board and the target device is measured, and the algorithm records the mean signal strength at certain distances.

• Task 4.2: AUV localization with RSSI

Aim: Get locations of AUVs by RSSI localization.

Expected Outcome: Localize at least 3 AUVs with positional accuracy within $\pm 7 \times 10^{-2}$ m and achieve a detection speed of at least 10Hz.

Group member in charge: CHAN, Sze Sen

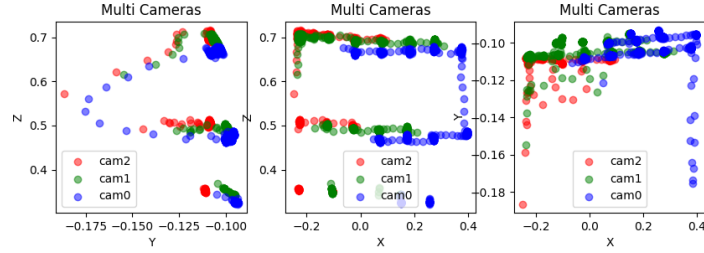


Figure 18: ArUco marker trajectory tracking in air showing consistent localization performance. The smooth path (blue line) with minimal scatter demonstrates accurate pose estimation without refractive distortion effects.

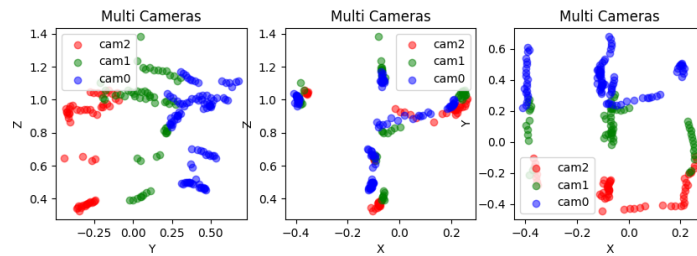


Figure 19: ArUco marker trajectory tracking underwater with the same distortion parameter in air. Showing increased localization variability due to refraction effects. The scattered position estimates highlight the need for refraction-corrected calibration parameters and robust multi-camera fusion algorithms.

Work:

2.3 Main Objective Evaluation & Discussion

3 Conclusion

TODO

Appendices

Appendix A - Project Schedule

Objective state-ments	Task	Group Member in charge	WK1 28/7	WK2 4/8	WK3 11/8	WK4 18/8	WK5 25/8	WK6 1/9	WK7 8/9	WK8 15/9	WK9 22/9	WK10 29/9	WK11 6/10	WK12 13/10	WK13 20/10	WK14 27/10	WK15 3/11	WK16 10/11	WK17 17/11
Simulation Environment Development																			
	Simulation Framework Setup	XU, Borong																	
	Noise Modeling Implementation	XU, Borong																	
	Validation and Testing of algorithms	XU, Borong																	
Camera Array Calibration																			
	Chessboard photo collection	CHAN, Size Sen																	
	Intrinsic Parameter Calibration	Both																	
	Extrinsic Parameter Calibration	Both																	
	Underwater Calibration Adjustment	Both																	
ArUco Tracking Implementation																			
	ArUco Localization Algorithm	Both																	
	Multi-Camera Fusion	XU, Borong																	
	AUV localization and results validation	Both																	
Sensor Fusion Implementation																			
	Extended Kalman Filter Implementation	XU, Borong																	

Appendix B - Budget

Table 2: List of budget

Items	Quantity	Cost per each (RMB¥)
WXSJ-H65HD V01 camera	4	¥112
S-YUE 7730V1 camera	1	¥108
UGREEN CM475 USB-C adapter	1	¥169
3D printed Camera holder	4	¥0*
ESP32-S3-WROOM-1 module board	1	¥0*
3D printed ESP32 holder	1	¥0*
M8 Screws 5pc	4	¥2.19
M8 T-Nut 10pc	2	¥1.6
M3 Screws and nuts 50pc	1	¥2.77
Aluminum extrusions 2020 1000mm	10	¥38.75
Aluminum extrusions 4040 1000mm	10	¥62.75
Calibration boards	5	¥0*
Contingency Budget (15%)		¥212.81
Total (including contingency)		¥1631.54

Budget Justification:

- **No-cost items (*):** 3D printed camera holders and calibration boards are provided by HKUST's Center for Engineering Education and Research Innovation (CEERI) through existing laboratory resources. This allocation supports academic research initiatives without additional expenditure, enabling access to professional-grade prototyping facilities.
- **Camera selection:** Five cameras (4× WXSJ-H65HD + 1× S-YUE) provide optimal cost-performance ratio at ¥110/camera average, offering 1280×960 resolution with OpenCV compatibility for real-time processing at 10 Hz.
- **Contingency allocation:** 15% contingency (¥212.81) covers potential hardware failures, additional calibration materials, or unexpected software licensing costs, ensuring project completion within budget constraints.
- **Cost optimization:** Total budget of ¥1418.73 represents significant savings compared to commercial AUV localization systems (typically ¥10,000+), achieving research-grade precision at educational price points while maintaining scalability for future multi-AUV experiments.

Appendix C - Meeting minutes

Meeting 1

- Date: 8/7/2025
- Time: 18:00
- Location: Zoom meeting
- Attendees: CHAN, Sze Sen, XU, Borong
- Absent: None
- Minutes taken by: CHAN, Sze Sen
 - Prof. Huan YIN introduces the background information of this project
 - We start planning the main objectives
 - Prof. YIN also states some goals we have to achieve, and can challenge
 - Primary Goals:
 - * Find the x-y coordinates of AUV
 - * Find the z coordinates of AUV
 - * Find the velocity of the AUV
 - Challenges:
 - * Solve the location shift between cameras during localization
 - * Localize the AUV with a single camera
 - * Scan the AUV when part of the ArUco tag has been blocked

Table 3: Action Items for Next Meeting

Action Item to be completed	By when	By whom
Visit CKSRI for setup inspection	13/7	CHAN, Sze Sen
Simulation Framework Setup	31/7	XU, Borong
Literature reviews on different localization methods	31/7	All

- Next Meeting: 5/8/2025, 19:30, Zoom meeting

Meeting 2

- Date: 5/8/2025
- Time: 19:30
- Location: Zoom meeting
- Attendees: CHAN, Sze Sen, XU, Borong
- Absent: None
- Minutes taken by: CHAN, Sze Sen
 - We come up with all the main objectives of the project
 - We all agree to use 5 cameras for localization, and to try applying IMU odometry as a challenge part later for the project
 - We plan the experiment of camera calibration and ArUco markers localization
 - The experiment will be held on 6/8 and 14/8, CHAN, Sze Sen will go to CKSRI for the hardware part, while XU, Borong will work on the software part online
 - Prof. YIN suggests the settings for capturing images of ArUco markers localization

Table 4: Action Items from the Previous Meeting

Action Item to be completed	By when	By whom	Status
Visit CKSRI for setup inspection	13/7	CHAN, Sze Sen	Completed
Simulation Framework Setup	31/7	XU, Borong	Completed
Literature reviews on other localization methods	31/7	All	Completed

Table 5: Action Items for Next Meeting

Action Item to be completed	By when	By whom
Image collection	14/8	CHAN, Sze Sen
Algorithms for intrinsic and extrinsic calibrations	14/8	XU, Borong
ArUco localization algorithm	22/8	All

- Next Meeting: 2/9/2025, 14:30, HKUST CKSRI

Meeting 3

- Date: 2/9/2025
- Time: 14:30
- Location: HKUST CKSRI
- Attendees: CHAN, Sze Sen, XU, Borong
- Absent: None
- Minutes taken by: CHAN, Sze Sen
 - The errors of intrinsic and extrinsic calibrations are $\pm 7.1 \times 10^{-2}$ m and $\pm 2.2 \times 10^{-2}$ m
 - Prof. YIN suggests reduce the errors within $\pm 2 \times 10^{-2}$ m
 - We plan to continue the lab experiment after finishing the proposal report
 - Prof. YIN also suggests several papers that could be referred to in the proposal report

Table 6: Action Items from the Previous Meeting

Action Item to be completed	By when	By whom	Status
Image collection	14/8	CHAN, Sze Sen	Completed
Algorithms for intrinsic and extrinsic calibrations	14/8	XU, Borong	Completed
ArUco localization algorithm	22/8	All	Completed

Table 7: Action Items for Next Meeting

Action Item to be completed	By when	By whom
Proposal Report Introduction	26/9	CHAN, Sze Sen
Proposal Report Methodology	26/9	XU, Borong
Finalize the proposal report	29/9	All

- Next Meeting: 8/10/2025, 14:30, HKUST CKSRI

Meeting 4

- Date: 8/10/2025
- Time: 14:30
- Location: HKUST CKSRI
- Attendees: CHAN, Sze Sen, XU, Borong
- Absent: None
- Minutes taken by: XU, Borong
 - We completed calibration of the camera's intrinsic parameters for single-camera localization in air, designing and mounting the camera array under the experimental water tank
 - We also designed a new method for camera calibration using ChArUco boards to improve the accuracy of cross-camera views
 - We are now facing difficulties with the theoretical model of multi-camera calibration, which requires further literature review, and the gap between in-air and underwater cross-medium environments needs validation through underwater experiments
 - Prof. YIN provides feedback on refining the calibration approach, suggested additional references for modeling cross-medium distortion, and also suggested implying RSSI localization as an assisting method
 - We all agree to explore RSSI as an assisting localization method, model cross-medium light refraction, and conduct initial underwater experiments

Table 8: Action Items from the Previous Meeting

Action Item to be completed	By when	By whom	Status
Proposal Report Introduction	26/9	CHAN, Sze Sen	Completed
Proposal Report Methodology	26/9	XU, Borong	Completed
Finalize the proposal report	29/9	All	Completed

Table 9: Action Items for Next Meeting

Action Item to be completed	By when	By whom
Literature review on RSSI for localization	22/10	CHAN, Sze Sen
Develop algorithm for cross-medium light refraction modeling	22/10	XU, Borong
Prepare and conduct initial underwater experiments	29/10	All

- Next Meeting: 5/11/2025, 14:30, HKUST CKSRI

Meeting 5

- Date: 5/11/2025
- Time: 14:30
- Location: HKUST CKSRI
- Attendees: CHAN, Sze Sen, XU, Borong
- Absent: None
- Minutes taken by: XU, Borong
 - We completed calibration of the camera's intrinsic parameters for single-camera localization in both air and underwater, calibration of extrinsic parameters for the new camera array and mounting under the water tank
 - We also developed and implemented algorithms to localize ArUco marks
 - The literature review on the implementation of RSSI as an assisting localization method was conducted
 - We are now facing difficulties with underwater experiments, which reveal a gap between in-air and underwater intrinsic calibration, and a suitable Python environment for the RSSI system requires further research
 - Prof. YIN provides feedback on addressing the calibration gap through additional modeling and suggested references for RSSI implementation and Python environments
 - We all agree to investigate RSSI implementation through further literature reviews, use mathematical models to optimize non-linear point projection from 2D to 3D, conduct more underwater experiments, and develop a Python environment showing the relationship between signal strength and AUV position

Table 10: Action Items from the Previous Meeting

Action Item to be completed	By when	By whom	Status
Literature review on RSSI for localization	22/10	CHAN, Sze Sen	Completed
Develop algorithm for cross-medium light refraction modeling	22/10	XU, Borong	Completed
Prepare and conduct initial underwater experiments	29/10	All	Completed

Table 11: Action Items for Next Meeting

Action Item to be completed	By when	By whom
Investigate implementation of RSSI through literature reviews	20/11	CHAN, Sze Sen
Use mathematical models to optimize non-linear point projection	20/11	XU, Borong
Conduct more underwater experiments to test the algorithm	27/11	All
Develop Python environment for signal strength and AUV position	27/11	All

- Next Meeting: 3/12/2025, 14:30, HKUST CKSRI

Meeting 6

- Date: 3/12/2025
- Time: 14:30
- Location: HKUST CKSRI
- Attendees: CHAN, Sze Sen, XU, Borong
- Absent: None
- Minutes taken by: CHAN, Sze Sen
 - The development of the Python environment for signal strength and AUV position is still in progress since further investigation on the module board given by Prof. YIN is required
 - Prof. YIN mentions that he would leave HKUST, so the supervisor of this project would be changed to Prof. Fumin ZHANG
 - We plan to continue the lab experiment after finishing the progress report and midterm poster
 - Prof. YIN suggests that we should include more about the gaps of traditional localization methods and our plan for the remaining parts of this project in the progress report, and add more images instead of much text to draw attention in the midterm poster

Table 12: Action Items from the Previous Meeting

Action Item to be completed	By when	By whom	Status
Investigate implementation of RSSI through literature reviews	20/11	CHAN, Sze Sen	Completed
Use mathematical models to optimize non-linear point projection	20/11	XU, Borong	Completed
Conduct more underwater experiments to test the algorithm	27/11	All	Completed
Develop Python environment for signal strength and AUV position	27/11	All	In progress

Table 13: Action Items for Next Meeting

Action Item to be completed	By when	By whom
Progress Report Objective Statement Execution	2/1	CHAN, Sze Sen
Progress Report Introduction and Methodology Overview	2/1	XU, Borong
Midterm Postser	8/1	All
Finalize the Progress Report	9/1	All

- Next Meeting: 13/1/2026, 15:00, HKUST CKSRI

Meeting 7

- Date: 13/1/2026
- Time: 15:00
- Location: HKUST CKSRI
- Attendees: CHAN, Sze Sen, XU, Borong
- Absent: None
- Minutes taken by: CHAN, Sze Sen
 - We discuss our division of work for the presentation in the midterm progress day on 23/1: CHAN, Sze Sen present the overview and objectives part; XU, Borong present the methodology and progress part
 - Prof. ZHANG suggests to use Arduino environment instead of Python for RSSI localization
 - We plan to test the reception of the signal strength of the ESP32 board with a simple algorithm in the Arduino environment, which only prints out the RSSI values of the target device
 - Prof. ZHANG also suggests that we can use a device such as a phone as a target device for reception testing

Table 14: Action Items from the Previous Meeting

Action Item to be completed	By when	By whom	Status
Progress Report Objective Statement Execution	2/1	CHAN, Sze Sen	Completed
Progress Report Introduction and Methodology Overview	2/1	XU, Borong	Completed
Midterm Poster	8/1	All	Completed
Finalize the Progress Report	9/1	All	Completed

Table 15: Action Items for Next Meeting

Action Item to be completed	By when	By whom
Preparation of Midterm Progress Day	23/1	All
Testing of reception of the signal strength of the ESP32 board	2/2	XU, Borong
Develop Arduino environment for signal strength and AUV position	11/2	CHAN, Sze Sen

- Next Meeting: 11/2/2026, 14:00, HKUST CKSRI

Meeting 8

- Date: 11/2/2026
- Time: 15:00
- Location: HKUST CKSRI
- Attendees: CHAN, Sze Sen, XU, Borong
- Absent: None
- Minutes taken by: XU, Borong
 - We designed an RSSI localization method and investigated the relationship between RSSI value and the distance between the target device and signal sensor.
 - We also investigated the calibration algorithm in OpenCV to apply the non-linear calibration.
 - We are now facing difficulties with significant errors in distance estimation, the signal from the target device is unstable, and the OpenCV calibration algorithm requires major modification for the non-linear calibration.
 - Prof. ZHANG provides feedback on improving RSSI data collection setup and signal stability, and suggested further detailed analysis of OpenCV non-linear calibration algorithms.
 - We all agree to increase the amount and range of signal data collection, conduct more reviews on the causes of signal instability, perform future investigations and detailed analyses of OpenCV camera calibration algorithms, and combine the non-linear calibration algorithm with the localization algorithm.

Table 16: Action Items from the Previous Meeting

Action Item to be completed	By when	By whom	Status
Preparation of Midterm Progress Day	23/1	All	Completed
Testing of reception of the signal strength of the ESP32 board	2/2	XU, Borong	Completed
Develop Arduino environment for signal strength and AUV position	11/2	CHAN, Sze Sen	Completed

Table 17: Action Items for Next Meeting

Action Item to be completed	By when	By whom
Increase the amount and range of signal data collection	25/2	CHAN, Sze Sen
More review on the causes of signal instability	25/2	XU, Borong
Detailed analysis of OpenCV camera calibration algorithms	1/3	All
Combine the non-linear calibration algorithm with the localization algorithm	1/3	All

- Next Meeting: 24/3/2026, 16:30, HKUST CKSRI

Appendix D - Group Members' Contribution

Individual Contributions:

- **XU, Borong:** Led algorithmic development including simulation framework, camera calibration, ArUco tracking, mechanical structure design, and sensor fusion implementation. Contributed significantly to technical documentation and mathematical derivations.

-
- **CHAN, Sze Sen:** I was responsible for hardware integration. To do this, I visited the HKUST CKSRI lab at the beginning of the project to inspect everything initially and ensure that the five-camera array was mounted correctly under the water tank we were using for our experiment, so that the overlapping fields of view from the cameras could provide optimal coverage.

For Objective 1, I contributed to system specification and validation testing of the Python simulation environment for camera calibration.

For Objective 2, I had to collect a large amount of data by capturing hundreds of chessboards and images in both air and underwater for camera calibration. I tried different pattern sizes and detection algorithms for the corners of these patterns to get the best results and minimize reprojection errors. For the experimental setup, I prepared and participated in initial and follow-up underwater tests, addressing challenges such as refraction distortions by validating air-water calibration gaps through hands-on trials and adjusting camera positions for better accuracy.

For Objective 3, I also helped with validating the system by testing the ArUco marker detection in real-life scenarios. I found problems with how it worked on cameras and optimized the performance in order to achieve 10Hz detection speed.

I led the documentation writing of this project. I prepared most of the meeting minutes and attached them to the report to fulfil the requirement. I prepared the introduction part of the proposal report. In this part, I stated the engineering problem of traditional localization methods. I also made literature reviews on existing solutions, including acoustic localization, magnetic localization, and optical localization. In the progress report, I added details of the objective statement execution part, including images and stating the process and challenges of the tasks. For both reports, I have to ensure that they meet all the requirements of the guidelines, and the format of writing from both members is unified. I also designed the midterm poster by including the main points of the progress report and figures we obtained during the experiment.

I coordinated laboratory resources by scheduling lab access, procuring components like cameras and adapters within budget, and ensuring proper experimental procedures, such as safety protocols during water tank operations.

Throughout, because of the suggestion from Prof. YIN, I conducted literature reviews on alternative localization methods, including RSSI as an assisting technique, investigating its implementation through further readings and suggesting integrations to enhance optical tracking robustness under occlusions. This practical focus complemented the algorithmic work, bridging simulation to real-world validation.

Appendix E - Deviation from the proposal and progress report and supporting reason

At the beginning, we try to apply optical localization only to localize the AUVs. However, once we try in the lab environment with less light, the accuracy of optical localization is limited. Therefore, we find RSSI localization as an assistant to overcome an insufficient light environment.

Appendix F - Monthly Reports

Monthly Report for ECE FYP/FYT

Project Code:	YH02	Supervisor(s):	Yin Huan
Project Title:	Development of a Multi-Camera Vision-Based Localization System for Tracking Multiple Micro Autonomous Underwater Vehicles (AUVs)		
Group Member(s):	1) CHAN, Sze Sen 2) XU, Borong		
Reporting Period:	Report #1 ☒ Oct (Fall) Report #2 ☐ Nov (Fall) Report #3 ☐ Feb (Spring) (please attach Reports #1-2 to the Progress Report to be submitted in Jan) (please attach Reports #3 to the Final Report to be submitted in Apr)		
Progress Report: ● List the work completed in this reporting period. ● Identify the major difficulties encountered. ● Comment on the overall progress.	Work completed: ● Calibrated the camera's intrinsic parameter for single-camera localization in the air. ● Designed the camera array and mounted it under the experimental water tank. ● Designed a new method for camera calibration using ChArUco boards to improve the accuracy of cross-camera views. Major difficulties: ● The theoretical model of multi-camera calibration is yet to be confirmed, requiring more literature review. ● The gap between an in-air environment and an underwater cross-medium environment is yet to be confirmed. Underwater experiments need to be conducted. Overall progress: ● Current progress is matching our goals.		
Future Plan: ● Write down the working plan for the next reporting period.	● Discover the implementation of RSSI as an assisting localization method through literature reviews. ● Model the cross-medium light refraction and develop an algorithm to fit the distortion. ● Conduct underwater experiments.		
Group Representative's Signature:			

(Version 2020-10)

Monthly Report for ECE FYP/FYT

Project Code:	YH02	Supervisor(s):	Yin Huan
Project Title:	Development of a Multi-Camera Vision-Based Localization System for Tracking Multiple Micro Autonomous Underwater Vehicles (AUVs)		
Group Member(s):	1) CHAN, Sze Sen 2) XU, Borong		
Reporting Period:	Report #1 ☒ Oct (Fall) Report #2 ☒ Nov (Fall) Report #3 ☐ Feb (Spring) (please attach Reports #1-2 to the Progress Report to be submitted in Jan) (please attach Reports #3 to the Final Report to be submitted in Apr)		
Progress Report: ● List the work completed in this reporting period. ● Identify the major difficulties encountered. ● Comment on the overall progress.	Work completed: ● Calibrated the camera's intrinsic parameters for single-camera localization in the air and under water. ● Calibrated the camera's extrinsic parameters for the new camera array and mount the camera array under the water tank. ● Developed and implemented algorithms to localize ArUco marks. ● Literature review on implementation of RSSI as an assisting localization method. Major difficulties: ● Underwater experiments revealed a gap between in-air and underwater intrinsic calibration. ● A suitable Python environment is yet to be confirmed for the RSSI system, requiring further research. Overall progress: ● Current progress is aligned with our goal, though slightly behind.		
Future Plan: ● Write down the working plan for the next reporting period.	● Investigate the implementation of RSSI as an assisting localization method through literature reviews. ● Use mathematical models to optimize the non-linear point projection from 2D point plane to 3D object point. ● Conduct more underwater experiments to test the algorithm. ● Develop a Python environment showing the relationship between signal strength and position of AUV		
Group Representative's Signature:			

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Monthly Report for ECE FYP/FYT

Project Code:	YH02	Supervisor(s):	Zhang Fumin
Project Title:	Development of a Multi-Camera Vision-Based Localization System for Tracking Multiple Micro Autonomous Underwater Vehicles (AUVs)		
Group Member(s):	1) CHAN, Sze Sen 2) XU, Borong		
Reporting Period:	Report #1 ☐ Oct (Fall) Report #2 ☐ Nov (Fall) Report #3 ☒ Feb (Spring) (please attach Reports #1-2 to the Progress Report to be submitted in Jan) (please attach Reports #3 to the Final Report to be submitted in Apr)		
Progress Report: ● List the work completed in this reporting period. ● Identify the major difficulties encountered. ● Comment on the overall progress.	Work completed: ● Designed RSSI-based localization method. ● Investigated the relationship between RSSI value and distance between signal emitter and receiver. ● Investigate the calibration algorithm in OpenCV to apply the non-linear calibration. Major difficulties: ● The distance estimation has significant errors. The setup of data collection has to be modified. ● Signal from the emitter is unstable. Experiment environment or choice of signal emitter has to be adjusted. ● The OpenCV calibration algorithm includes a backward mapping and an updated intrinsic parameter for localization, major modification should be made for the non-linear calibration algorithm. Overall progress: ● Current progress is matching our goals.		
Future Plan: ● Write down the working plan for the next reporting period.	● Increase the amount and range of signal data collection. ● More review on the causes of signal instability. ● Future investigation and detailed analysis of OpenCV camera calibration algorithms. ● Combine the non-linear calibration algorithm with the localization algorithm.		
Group Representative's Signature:			

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